US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250271155
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	Mutter; David S. et al.

Roof Vent Flashing with a Fire and Ember Barrier Border

Abstract

A vent as described herein provides for a fire and ember barrier border incorporated around the perimeter of vent flashing and/or sub-flashing. In the event that a portion of vent flashing separates from the underlayment to which it was originally attached, it creates a secondary air path between the deck and the flashing and the fire and ember barrier border will prevent fire and embers from entering the secondary air path and ultimately the building.

Inventors: **Mutter; David S. (Santa Rosa, CA), Daniels; Gregory S. (Santa Rosa, CA), Daniels, II; William B. (Las Vegas, NV)**

Applicant: **O'Daniels, LLC.** (Henderson, NV)

Family ID: **1000007828889**

Assignee: **O'Daniels, LLC. (Henderson, NV)**

Appl. No.: **18/589412**

Filed: **February 27, 2024**

Publication Classification

Int. Cl.: **F24F7/02** (20060101); **A62C2/06** (20060101); **A62C3/02** (20060101)

U.S. Cl.:

CPC **F24F7/02** (20130101); **A62C2/06** (20130101); **A62C3/0214** (20130101);

Background/Summary

FIELD OF THE INVENTIONS
[0001] The inventions described below relate to the field of vents for building ventilation.

BACKGROUND OF THE INVENTIONS

[0002] Energy efficiency is a serious consideration in every building. Residential buildings require ventilation systems to minimize heat and moisture buildup in the uninhabited spaces within the building such as roofs, soffits, attics, basements and crawl spaces. In areas with extreme weather, rain, fire and embers, these undesirable elements are often driven into and through ventilation systems causing damage to the building, sometimes completely destroying it. In many circumstances the undesirable elements enter the ventilation system around and under flashing and/or sub-flashing that was either installed incorrectly, inexpertly, or has aged badly and separated from the underlayment.

SUMMARY

[0003] The devices and methods described below provide for a fire and ember barrier border incorporated around the entire perimeter of vent flashing and sub-flashing. In the event that a portion of vent flashing separates from the underlayment to which it was originally attached, the separation creates a secondary air path between the underlayment and the flashing and the fire and ember barrier border will prevent fire and embers from entering the secondary air path and ultimately the building.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a simplified perspective view of a building with ventilation through the roof, foundation and gable.
[0005] FIG. **2** is a perspective view of a low profile roof vent with a fire and ember barrier border suitable for use on the building of FIG. **1**.
[0006] FIG. **3** is a cross section view of the vent of FIG. **2** taken along A-A.
[0007] FIG. **4** is a top view of the vent of FIG. **2**.
[0008] FIG. **5** is a bottom view of the vent of FIG. **2**.
[0009] FIG. **6** is a perspective view of a flat tile roof with a shape matching vent.
[0010] FIG. **7** is a perspective view of an S-type tile roof with a shape matching vent.
[0011] FIG. **8** is a cross section view of the vent of FIG. **6** taken along B-B.
[0012] FIG. **9** is a cross section view of the vent of FIG. **7** taken along C-C.
[0013] FIG. **10** is a top perspective view of an M-type secondary vent.
[0014] FIG. **11** is a bottom perspective view of the vent of FIG. **10**.
[0015] FIG. **12** is a side view of the vent of FIG. **10**.
[0016] FIG. **13** is a perspective view of a passive foundation vent with a fire and ember barrier border.
[0017] FIG. **14** is a cross-section view of the passive foundation vent of FIG. **13** with a fire and ember barrier border taken along E-E.
[0018] FIG. **15** is a top view of a primary vent with a fire and ember resistant border.
[0019] FIG. **16** is a side view of the primary vent with a fire and ember resistant border of FIG. **13**.
[0020] FIG. **17** is an end view of the primary vent with a fire and ember resistant border of FIG. **13**.
[0021] FIG. **18** is a bottom view of the primary vent with a fire and ember resistant border of FIG. **13**.

DETAILED DESCRIPTION OF THE INVENTIONS

[0022] Buildings such as building **1** of FIG. **1** have one or more ventilation apertures, such as apertures **2**A-**2**H, through the roof deck **1**D, apertures under the eaves, aperture **3** in the gable, or apertures **4**A and **4**B to permit ventilation of the basement, crawlspace, cellar, and or undercroft of the building. The ventilation apertures on the roof may be arranged with some apertures higher and or lower on the roof to optimize passive ventilation of the building. For example, the upper or upslope apertures **5** such as apertures **2**A-**2**D are oriented high on the roof, upslope from the lower or downslope apertures **6**, such as apertures **2**E-**2**H. This orientation permits passive air circulation, convection, with outside air **7** entering the lower apertures and exiting the upper apertures as exhaust air **7**X carrying heat and moisture from within the attic. Lower apertures may be located through the roof deck or alternatively or in addition to having lower apertures on the roof deck one or more additional eave apertures **8** may be included under the eaves **1**E of the building.

[0023] Any building ventilation aperture, such as the ventilation apertures **2**A-**2**H, **3**, **4**A and **4**B, are protected by any suitable vent cover such as the low profile vent **10** of FIGS. **2** and **3** which is suitable for use with shingle, metal, slate or shake roofs, flat tile vent **11** of FIGS. **6** and **8** which is suitable for use with flat clay or concrete roof tiles, S-type vent **12** of FIGS. **7** and **9** which is suitable for use with S-type clay or concrete roof tiles, M-type vent **13** of FIGS. **10**, **11** and **12** which is suitable for use with S-type clay or concrete roof tiles or foundation vent **15** which is suitable for foundation uses to ventilate basements and crawl spaces.

[0024] Two-piece vents are composed of a primary vent and a secondary vent in fluid communication. Flat tile vent **11**, S-type vent **12** and M-type vent **13** are secondary vents and are illustrated in FIGS. **6** through **12**. Primary vent **14** is illustrated in FIGS. **15**, **16**, **17** and **18**. A primary vent, such as vent **14**, is located between any suitable two-piece vent and a ventilation aperture through the roof deck such as ventilation apertures **2**A-**2**H and enables fluid communication between a ventilation aperture and the secondary vent as illustrated in FIGS. **8** and **9**.

[0025] The low profile vent **10** of FIGS. **2**, **3**, **4** and **5** has vent flashing **10**F with a fire and ember barrier border **10**B secured to the flashing and extending around the perimeter **10**P of the flashing. The vent flashing **10**F and fire and ember barrier border **10**B are secured to the roof deck **1**D with the vent oriented with vent opening **16** in fluid communication with a ventilation aperture through the roof decking such as one of ventilation apertures **2**A-**2**H. Vent flashing **10**F as well as fire and ember barrier border **10**B are secured to the deck using any suitable material, technique or combination of materials and techniques including without limitation adhesives, caulking and or fastening with staples, nails, screws, brads or other suitable components. Vent **10** includes a downslope vent opening **17** in fluid communication with vent cavity **18** as well as vent opening **16**. Vent opening **16** is covered by screen **16**S which is any suitable ventilation screen. Screen **16**S and fire and ember barrier border **10**B may be a suitable ventilation screen such as ¼″ steel mesh although any suitable material and mesh size may be used such as ⅛″ mesh made of stainless steel, brass, copper, plastic, intumescent material or other.

[0026] Similarly, foundation vent **15**, illustrated in FIGS. **13** and **14**, is directly secured into any suitable ventilation aperture such as foundation vent apertures **4**A and **4**B illustrated in FIG. **1**. Foundation vent **15** includes vent cavity **15**C in fluid communication between the building foundation space and the building exterior. Flashing **15**F of foundation vent **15** has a perimeter **15**P and fire and ember barrier border **15**B is secured to the flashing **15**F completely surrounding the perimeter **15**P of flashing **15**F. Flashing **15**F and fire and ember barrier border **15**B are secured to the building using any suitable material, technique or combination of materials and techniques including without limitation adhesives, caulking and or fastening with staples, nails, screws, brads or other suitable components. Fire and ember barrier border **15**B may be a suitable ventilation screen such as ¼″ steel mesh although any suitable material and mesh size may be used such as ⅛″ mesh made of stainless steel, brass, copper, plastic, intumescent material or other.

[0027] Primary vent **14** is formed of flashing **14**F with

[0028] ventilation opening **19** therethrough. Ventilation opening **19** is oriented to be in fluid communication with a suitable ventilation aperture such as ventilation apertures **2**A-**2**H, **3**, **4**A and **4**B. Primary vent **14** includes any suitable insect/vermin and debris barrier covering ventilation opening **19**, such as screen **14**S, and fire and ember barrier border **14**B. Flashing **14**F has a perimeter **14**P, fire and ember barrier border **14**B which is attached to the flashing, completely surrounds the perimeter **14**P. Primary vent **14** is installed on a roof deck, such as deck **1**D of FIG. **1**, by fastening the flashing **14**F and fire and ember barrier border **14**B to the deck using any suitable material, technique or combination of materials and techniques including without limitation adhesives, caulking and or fastening with staples, nails, screws, brads or other suitable components. Primary vent **14** is oriented with ventilation opening over a ventilation opening or aperture, such as ventilation apertures **2**A-**2**H, cut through the deck.

[0029] Any of vents **10**, **11**, **12**, **13**, **14**, and or **15** may include optional filter mesh **9**, as illustrated in FIGS. **3** and **14**, to provide additional protection against debris and or embers transiting the vent system. The optional filter mesh is a flame-resistant interwoven mesh which may be any suitable material such as stainless steel. In a preferred configuration, the optional filter mesh is stainless steel wool made from alloy type AISI 434 stainless steel which forms a pad approximately ¼″ thick. The optional filter mesh may be secured to screen any of the skeletons, flashing or other suitable support using any suitable material or technique, including without limitation adhesives, caulking, welding, fastening, wrapping, stitching and the like.

[0030] Referring now to FIGS. **10**, **11** and **12**, M-type vent **13** has a skeleton **13**S parallel to cap **13**C. The orientation of skeleton **13**S and cap **13**C create a downslope vent opening **20** in fluid communication with vent cavity **21** formed between skeleton **13**S and cap **13**C. Skeleton **13**S includes secondary ventilation openings **22**A, **22**B and **22**C.

[0031] Flat tile vent **11** of FIGS. **6** and **8** has a skeleton **11**S parallel to cap **11**C. The orientation of skeleton **11**S and cap **11**C create a downslope vent opening **26** in fluid communication with vent cavity **27** formed between skeleton **11**S and cap **11**C. Skeleton **11**S includes a secondary ventilation opening **28** in fluid communication with vent cavity **27** and downslope vent opening **26** as well as vent opening **19** of primary vent **14**.

[0032] S-type vent **12** of FIGS. **7** and **9** has a skeleton **12**S parallel to cap **12**C. The orientation of skeleton **12**S and cap **12**C create a downslope vent opening **32** in fluid communication with vent cavity **33** formed between skeleton **12**S and cap **12**C. Skeleton **12**S includes first and second secondary ventilation openings **34**A and **34**B which permit airflow from the attic through the primary vent and the secondary vent. Ventilation aperture **2**A is in fluid communication with primary vent opening **19** which is in fluid communication with secondary vent openings **34**A and **34**B which are in fluid communication with vent cavity **33** which is in fluid communication with downslope vent opening **32**.

[0033] While the preferred embodiments of the devices and methods have been described in reference to the environment in which they were developed, they are merely illustrative of the principles of the inventions. The elements of the various embodiments may be incorporated into each of the other species to obtain the benefits of those elements in combination with such other species, and the various beneficial features may be employed in embodiments alone or in combination with each other. Other embodiments and configurations may be devised without departing from the spirit of the inventions and the scope of the appended claims.

Claims

1. A ventilation system for the roof of a building comprising: a primary vent having flashing with a ventilation opening therethrough, the flashing having perimeter with a fire and ember barrier border secured to and surrounding the perimeter of the flashing, the ventilation opening in fluid communication with a ventilation aperture through the roof; a debris barrier covering the ventilation opening; and a secondary vent with a downslope ventilation opening in fluid communication with a vent cavity and one or more secondary vent openings in fluid communication with the ventilation opening of the primary vent.

2. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ¼″ steel mesh.

3. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ⅛″ steel mesh.
4. The ventilation system of claim 2 wherein the ¼″ steel mesh is stainless steel.
5. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ¼″ brass mesh.
6. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ¼″ copper mesh.
7. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ¼″ plastic mesh.
8. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ¼″ intumescent mesh.
9. The ventilation system of claim 3 wherein the ⅛″ steel mesh is stainless steel.
10. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ⅛″ brass mesh.
11. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ⅛″ copper mesh.
12. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ⅛″ plastic mesh.
13. The ventilation system of claim 1 wherein the fire and ember barrier border and the debris barrier are ⅛″ intumescent mesh.
14. the ventilation of claim 1 further comprising: A filter mesh in the vent cavity.
15. A ventilation system comprising: a low profile vent having flashing with a ventilation opening therethrough, the flashing having a perimeter with a fire and ember barrier border secured to and surrounding the perimeter of the flashing and a debris barrier covering the ventilation opening.
16. The ventilation system of claim 15 wherein the fire and ember barrier border and the debris barrier are ¼″ steel mesh.
17. The ventilation system of claim 15 wherein the fire and ember barrier border and the debris barrier are ⅛″ steel mesh.
18. A building ventilation system comprising: a foundation vent having a flashing operable to secure the vent to a building, the flashing having a perimeter; and a fire and ember barrier border secured to and surrounding the perimeter of the flashing.